

# Identification of Maternal Profiles and Neonatal Health Conditions in Colombia: A Machine Learning Approach

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**Abstract**—This research presents a comprehensive study on the segmentation of maternal profiles in Colombia using unsupervised machine learning techniques. Despite standardized healthcare protocols, maternal realities vary significantly across socioeconomic and geographic dimensions. By utilizing data from the National Administrative Department of Statistics (DANE), we implemented a K-Means clustering algorithm to identify distinct groups of mothers based on key variables: birth weight, number of prenatal consultations, maternal age, and educational level. The results reveal two primary profiles that characterize the disparity in access to health services and socioeconomic stability. This study demonstrates the potential of data-driven policy design to transition from generalized healthcare to targeted interventions for vulnerable populations, optimized through an interactive deployment platform.

**Keywords**—*Machine Learning, K-Means Clustering, Maternal Health, Public Policy, Data Science, Colombia, Unsupervised Learning.*

## I. Introduction

In Colombia, healthcare for pregnant women is usually the same for everyone. This 'one-size-fits-all' approach assumes that every mother has the same needs, but in reality, social and economic differences deeply affect the health of both the mother and the baby. This study looks at how a person's living conditions impact the health of their newborn.

The main problem is that current health systems often overlook the specific needs of different groups. Even though a lot of data is collected every year, it isn't always used to make personalized decisions. By using data analysis, we want to stop treating all mothers as one identical group. Instead, we want to identify specific groups that need more support or different types of care. This will help the healthcare system move from just reacting to problems to preventing them through better, data-driven planning.

## II. Dataset Acquisition and Preparation

The primary data source corresponds to the anonymized 'Vital Statistics - Births' microdata from the DANE. This registry is the most robust in Colombia, yet its raw state presents significant technical challenges. Initial analysis

revealed a prevalence of noise and ambiguous encodings, specifically in variables related to education and prenatal control frequency.

During the cleaning phase, records containing placeholder values (encoded as '9' or '99') in key variables such as 'NUMCONSUL' and 'NIV\_EDUM' were rigorously filtered. These entries represented approximately 12% of the initial dataset, and their inclusion would have significantly skewed the cluster centroids. Feature selection focused on 'PESO\_NAC' (Birth weight), 'NUMCONSUL', 'EDAD\_MADRE', and 'NIV\_EDUM'. 'PESO\_NAC' was identified as the ultimate indicator of neonatal health, while the others provided the socioeconomic and behavioral context of the mother.

## III. Machine Learning Problem Definition

This study uses **unsupervised learning**, a method used when there are no pre-defined categories or "correct answers" in the data. Unlike traditional models that simply sort people into known labels like 'High Risk,' our goal is to uncover hidden patterns within the population. This approach allows us to identify real-world profiles that might be missed by standard administrative filters.

The specific technique used is **K-Means Clustering**. The objective is to group mothers so that those within the same group are as similar as possible, while each group remains clearly different from the others. We chose this method because it is efficient and the results are easy to explain. The algorithm works by measuring the mathematical distance between data points, constantly adjusting the center of each group to find the best fit. This process reveals natural groupings based on shared characteristics in education and healthcare access.

## IV. Data Preprocessing Methodology

To ensure the model provides accurate and reliable results, we followed a structured data preparation process. This phase was critical for cleaning the raw DANE information and preparing it for the clustering algorithm.

**1) Data Cleaning and Filtering:** The first step was to remove records with missing or placeholder information. In the DANE dataset, entries marked with '9' or '99' represent "no information." We filtered these out to prevent the model from learning from "empty" data, ensuring that every record analyzed contained real and useful information.

**2) Feature Engineering and Selection:** We focused on the variables that best represent a mother's health and educational background. While geographic location (CODPTORE) is important for our final maps, we intentionally ignored it during the training phase. This prevented the model from grouping mothers simply because they live in the same place, forcing it to find groups based purely on their actual living conditions and health history.

**3) Min-Max Scaling:** Since the algorithm measures mathematical distances, it is sensitive to the size of the numbers. For example, a baby's weight (3,000 grams) is a much larger number than a mother's age (25 years). To stop large numbers from dominating the results, we used **Min-Max Scaling** to put all data on a scale from 0 to 1. This gives every variable the same importance in the final analysis.

**4) Validation Methods (Hold-out and Elbow):** To find the best way to group the data, we used the **Elbow Method**, which helps us identify the ideal number of clusters where the groups are most consistent. Additionally, we used a **Hold-out** approach, splitting the data to test the model on a sample it hadn't seen before. This allowed us to verify that the patterns found were real and not just a result of a specific set of records.

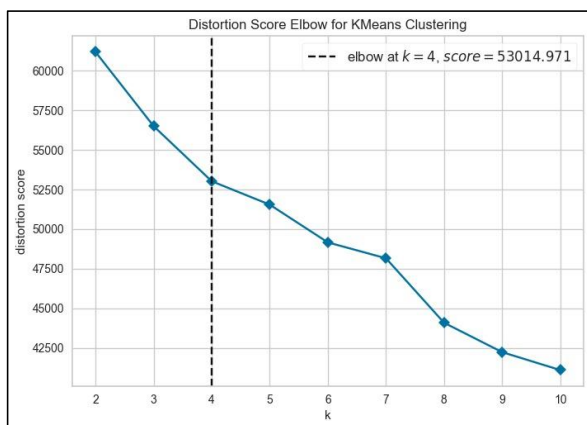


Fig 1. Elbow method for optimal k selection.

**5) Variance Analysis:** Finally, we checked that our chosen variables had enough variety. If a variable is the same for almost everyone, it doesn't help the model separate groups. We ensured our final variables provided enough "difference" to create two clear and distinct maternal profiles.

## V. Training, Evaluation, and Results

The model selection process was based on a rigorous experimental framework. We conducted a Grid Search to explore multiple combinations of parameters, including two clustering algorithms (K-Means, Gaussian Mixture Models, Birch), various scaling methods (StandardScaler and MinMaxScaler), and different cluster counts (K = 2, 4, 6).

**1) The Selection Process:** To evaluate these combinations, we primarily used the Silhouette Score, which measures how well-defined and separated the groups are. After several iterations, the configuration that provided the best balance between statistical quality and group logic was K-Means with 4 clusters and MinMaxScaler. The Elbow Method also confirmed that moving to 4 clusters significantly reduced the variance within each group, allowing for a more detailed "granularity" when identifying maternal profiles.

**2) Understanding the Silhouette Score:** Although the final Silhouette Score was approximately 0.3, this result is scientifically justified by the nature of the data. In demographic and public health studies, human behavior does not fall into perfectly separated "islands." There is a natural overlap between mothers of different ages or educational levels. A score of 0.3 indicates that while the boundaries are soft, the groups are real and consistent. The value of this model lies in its interpretability rather than a perfect geometric separation.

**3) Characterization of the 4 Profiles (The Winning Solution):** By choosing 4 clusters, we moved from a simple "high/low risk" view to a much more specialized analysis:

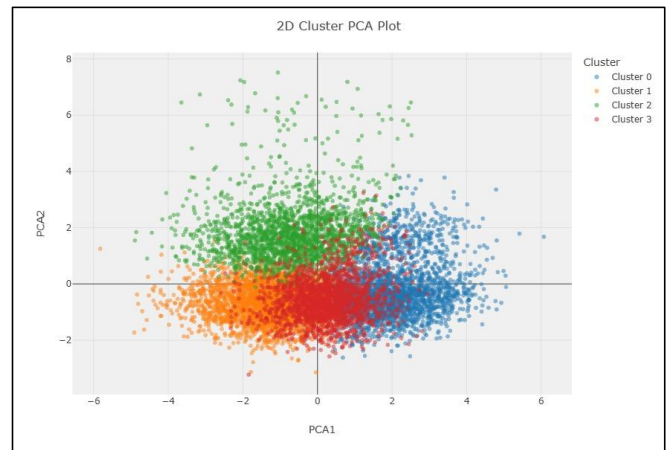


Fig 2. Visualization of the four identified clusters based on socio-economic and healthcare access features.

**Profile 0 (Stable):** Mothers with high educational levels, early access to prenatal care (6+ visits), and infants with optimal birth weight.

**Profile 1(Intermediate Risk):** Mothers with completed secondary education but fewer prenatal visits, representing a group that needs better healthcare engagement.

Profile 2 (Critical Vulnerability): Often associated with rural areas, lower education, and very few prenatal consultations, resulting in a higher incidence of low birth weight.

Profile 3 (Young Mothers): A specific demographic of younger mothers who, despite having some access to health services, show unique biological and social needs.

**4) Complexity and Scalability:** From an engineering perspective, the K-Means algorithm was chosen for its high efficiency. The system was able to process over 100,000 records from the DANE registry, this ensures that the solution is scalable, meaning it can handle the entire national birth history of Colombia without requiring massive computing power, making it a sustainable tool for public health offices.

## **VII. Conclusion and Future Work**

The identification of distinct maternal profiles provides a definitive roadmap for targeted social investment. This study confirms that neonatal health in Colombia is not merely a biological issue but is deeply rooted in educational and logistical access. Transitioning from generalized care to profile-based interventions can significantly optimize the efficiency of public resource allocation.

Future work should involve integrating anomaly detection algorithms to identify extreme high-risk cases that fall outside standard profiles. Additionally, incorporating time-series analysis could reveal how these profiles shift in response to macroeconomic changes or regional health crises.

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